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Data Analyst Intern Take-Home Assignment

Good to know before you start:

- This is designed to take about an hour of real thinking. There's no timer, do it in one sitting or spread it across the next couple of days, whenever you can give it an uninterrupted stretch.
- There are three self-contained parts. You don't need a shipping/logistics background for any of them, if a fact is missing, state your assumption and move on.
- We care more about how you get to an answer than whether it matches some “textbook” one. A short, clearly-reasoned response beats a long, hedged one.
- There's no scoring rubric we're holding you to, just use your own judgement on what a good answer looks like, the same way you would on the job.
- AI tools are welcome throughout. Just be specific about where and how you used them, see the note at the very end.

Part A: Root Cause Analysis (~20 min)

A customer's container was quoted an ETA of 2024-04-15 into Rotterdam. It actually arrived on 2024-04-22 (7 days late) and the customer wants to know why, since “ocean delays” is the explanation they're always given and they no longer trust it. You pull the booked routing and the actual tracking events for the container.

Booked routing (as promised to the customer):

Leg	Vessel (booked)	Vessel IMO	Departure (planned)	Arrival (planned)
1: Shanghai (CNSHA) → Singapore (SGSIN)	EVER GIVEN	9811000	2024-03-02	2024-03-09
2: Singapore (SGSIN) → Rotterdam (NLRTM)	MSC ANNA	9839430	2024-03-11	2024-04-15 (promised ETA)

Actual milestone events (from AIS + terminal tracking data):

Milestone Event	Vessel (actual)	Vessel IMO	Timestamp
Gate-in, Shanghai	—	—	2024-03-01
Loaded, Shanghai	EVER GIVEN	9811000	2024-03-02
Discharged, Singapore	EVER GIVEN	9811000	2024-03-10
Loaded, Singapore	COSCO PACIFIC	9904455	2024-03-17
Departed, Singapore	COSCO PACIFIC	9904455	2024-03-18
Arrived, Rotterdam	COSCO PACIFIC	9904455	2024-04-22 (actual)

Answer using the data, not intuition:

- What actually caused the 7-day delay, on-ocean schedule slippage, port congestion at Singapore, or a rollover onto a later vessel? Pick one and justify it.
- What specific detail in the two tables rules out each of the other two explanations?
- If you were monitoring shipments at scale, what signal would you check for automatically so this gets flagged before the customer has to ask?

Part B: Guesstimate (~25 min)

A mid-size manufacturer ships roughly 400 containers a month across its network. Today, the vessel ETAs they plan around come straight from carrier schedules, which are known to drift by a few days versus what actually happens.

Question:

"What is a reasonable order-of-magnitude estimate for the annual cost this company incurs because of inaccurate ETAs through things like safety stock, expedited freight, and port/warehouse detention or demurrage?"

We are not looking for a "correct" number. We're evaluating how you break an ambiguous, real-world question into something calculable. Please include:

- The 3–5 assumptions you're making, stated explicitly (you can invent reasonable numbers).
- A clear formula or logic tree connecting your assumptions to the final estimate.
- Your final range (an order of magnitude is fine, e.g. "$2–5M/year").
- One line on which single assumption your answer is most sensitive to, and why.

Part C: AI Output Sanity Check (~15 min)

Your company is integrating carrier and terminal track-and-trace data at scale, 50+ carriers and terminals feeding shipment milestone events into one system. A colleague pastes the paragraph below into Slack, an AI copilot's draft plan for how to approach the integration, and asks "can we just run with this?"

> *"To integrate carrier and terminal track-and-trace feeds at scale, we recommend building a single universal parser that maps every incoming feed - AIS pings, carrier APIs, terminal EDI messages, PDF schedules - into one common schema, since virtually all major carriers and terminals now expose structured, real-time APIs on similar standards. Because each feed is already validated by the carrier or terminal that produces it, we don't need a separate data-quality or reliability layer on our side - we can treat every incoming milestone event as equally trustworthy and pass it straight into the customer-facing ETA model. This lets us onboard all 50+ integrations within the current sprint and go live in two weeks."*

Answer briefly:

- List the specific claims or decisions in this paragraph you would push back on before the team commits to this plan, and why each one is a problem.
- Rewrite the two most important sentences so the plan is still useful but no longer makes those mistakes.

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One last thing: tell us about AI vs. your own effort

For each part (A, B, C), give us a couple of sentences on:

- Whether you used any AI tools, and for what, e.g. brainstorming, checking your math, drafting language, generating the whole answer.
- What was purely your own thinking versus what came from a tool.

Be specific rather than a blanket “used ChatGPT for this.” This doesn't count against you — we'd genuinely rather you use whatever tools you'd use on the job but a vague or missing answer here does.

Evaluation=

- Structured thinking: Can you break an ambiguous, multi-cause problem into something you can actually reason through, rather than jumping to a guess?
- Problem-solving under real constraints: Do your conclusions follow from the data/assumptions in front of you, not from what "sounds right"?
- Ownership: Did you push through all three parts to a complete, decisive answer, and are you honest and specific about where AI helped versus where it was their own thinking?

This exercise is designed to test the core skills required for this role. Given the number of submissions we received, we will be highly selective and are looking for candidates who go beyond simply getting the right answer and demonstrate A+ level thinking, first principle problem-solving, and ownership.

Submission

Send your submission to g@portcast.io. You are free to choose whichever format you want to share the submission in.

Disclaimer

This exercise is for skills assessment only and your submission will not be used by the company.